

The Bayesian Bridge Between Simple and Universal Kriging¹

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Kriging techniques are suited well for evaluation of continuous, spatial phenomena. Bayesian statistics are characterized by using prior qualified guesses on the model parameters. By merging kriging techniques and Bayesian theory, prior guesses may be used in a spatial setting. Partial knowledge of model parameters defines a continuum of models between what is named simple and universal kriging in geostatistical terminology. The Bayesian approach to kriging is developed and discussed, and a case study concerning depth conversion of seismic reflection times is presented.

KEY WORDS: spatial statistics, regionalized variables, Bayesian statistics, seismic depth conversion.

INTRODUCTION

In Omre (1987), a Bayesian approach to kriging was presented. The formulation can be considered an extension of the simple kriging formulation allowing for uncertainty in the specification of the expected function. Several arguments for using prior guesses on model parameters instead of introducing fake observations were forwarded.

In this paper, simple and universal kriging are presented in a Bayesian setting. Introduction to universal kriging theory can be found in Matheron (1971) and to Bayesian theory in Berger (1980). In the former, the introduction of random drift is made, but it is used mainly to demonstrate robustness of universal kriging. Note that if the expectation of a random function $[Z(x); x \in A]$ has the following form:

$$E[Z(x)] = \sum_{l=1}^L b_l f^l(x)$$

with regionalized variables $[f^l(x); x \in A]; l = 1, \dots, L$ known, the existing kriging theory provides two options. Either assume $b_l; l = 1, \dots, L$ as known

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and use simple kriging, or assume b_l ; $l = 1, \dots, L$ as unknown and use universal kriging. The intrinsic random function of order k is, practically spoken, identical to the universal kriging approach. The Bayesian approach makes room for partial knowledge of the coefficients, and defines simple and universal kriging as two extreme cases. Hence, a continuum of model formulations between simple and universal kriging is defined.

In the next section, some basic notation and relations are defined. In the third section, an outline of the estimation procedure for obtaining the spatial covariance is given. The fourth section contains the definition of the Bayesian kriging relations and their solutions. In the following section the definition of the posterior parameters is presented, and thereafter some special cases are discussed. The seventh section contains an example of use on depth conversion of seismic reflection times for petroleum exploration. Some theoretical aspects are discussed in greater detail in the appendix.

BASIC NOTATION AND RELATIONS

Let the regionalized variable under study be denoted $[z(x); x \in A]$, and let the underlying random function be

$$[Z(x); x \in A]$$

Consider a set of random variables

$$[B_l; l = 1, \dots, L]$$

with the two first moments known *a priori*:

$$E[B_l] = \mu_l \quad l = 1, \dots, L$$

$$\text{Cov}[B_l, B_k] = \sigma_{lk} \quad l, k = 1, \dots, L$$

Let the random function and the set of random variables be related by

- Expectation:

$$E[Z(x) | B_l; l = 1, \dots, L] = \sum_{l=1}^L B_l f^l(x)$$

with L a user specified integer constant; $[f^l(x); x \in A]$; $l = 1, \dots, L$ user specified regionalized variables; $[B_l; l = 1, \dots, L]$ defined as above.

- Spatial covariance:

$$\text{Cov}[Z(x'), Z(x'') | B_l; l = 1, \dots, L] = C_{Z|B}(x' - x'')$$

with $C_{Z|B}(\cdot)$ user specified spatial covariance function which can be estimated from observations.

Hence $[B_l; l = 1, \dots, L]$ can be interpreted as a guess, associated with uncertainty, of coefficients in the universal kriging formulation of the random function $[Z(x); x \in A]$. The covariance assumption entails that $[Z(x); x \in A]$ is second order stationary around its true expected function $[\sum_l b_l f^l(x); x \in A]$. This is an expression familiar to geostatisticians. Note also that the corresponding variogram function is:

$$\begin{aligned}\gamma_{Z|B}(x' - x'') &= \frac{1}{2} \text{Var} [Z(x') - Z(x'') \mid B_l; l = 1, \dots, L] \\ &= \text{Var} [Z(x) \mid B_l; l = 1, \dots, L] \\ &\quad - \text{Cov} [Z(x'), Z(x'') \mid B_l; l = 1, \dots, L] \\ &= C_{Z|B}(0) - C_{Z|B}(x' - x'')\end{aligned}\quad (1)$$

Recall the following expressions for conditional expectation and conditional covariances,

$$E[Y_1] = E[E[Y_1 \mid Y_3]]$$

$$\text{Cov}[Y_1, Y_2] = E[\text{Cov}[Y_1, Y_2 \mid Y_3]] + \text{Cov}[E[Y_1 \mid Y_3], E[Y_2 \mid Y_3]] \quad (2)$$

with Y_1, Y_2 , and Y_3 arbitrary random variables.

The definitions above and these expressions entail

$$\begin{aligned}\mu_Z(x) &= E[Z(x)] = E[E[Z(x) \mid B_l; l = 1, \dots, L]] \\ &= \sum_l \mu_l f^l(x); x \in A\end{aligned}$$

$$\begin{aligned}C_Z(x', x'') &= \text{Cov}[Z(x'), Z(x'')] \\ &= E[\text{Cov}[Z(x'), Z(x'') \mid B_l; l = 1, \dots, L]] \\ &\quad + \text{Cov}[E[Z(x') \mid B_l; l = 1, \dots, L], E[Z(x'') \mid B_l; \\ &\quad l = 1, \dots, L]]; \\ &= C_{Z|B}(x' - x'') + \sum_l \sum_k \sigma_{lk} f^l(x') f^k(x''); x', x'' \in A\end{aligned}\quad (3)$$

$$\begin{aligned}\gamma_Z(x', x'') &= \frac{1}{2} \text{Var}[Z(x') - Z(x'')] \\ &= E[\text{Var}[Z(x') - Z(x'') \mid B_l; l = 1, \dots, L]] \\ &\quad + \text{Var}[E[Z(x') - Z(x'') \mid B_l; l = 1, \dots, L]] \\ &= \gamma_{Z|B}(x' - x'') + \sum_l \sum_k \sigma_{lk} (f^l(x') - f^l(x'')) \\ &\quad \cdot (f^k(x') - f^k(x'')); x', x'' \in A\end{aligned}$$

$$\begin{aligned}
C_{ZB_k}(x) &= \text{Cov} [Z(x), B_k] \\
&= E[\text{Cov} [Z(x), B_k | B_l; l = 1, \dots, L]] \\
&\quad + \text{Cov} [E[Z(x) | B_l; l = 1, \dots, L], E[B_k | B_l; l = 1, \dots, L]] \\
&= \sum_l \sigma_{lk} f^l(x); k = 1, \dots, L; x \in A
\end{aligned}$$

Note that the marginal covariance and variogram function of $[Z(x); x \in A]$ are dependent on absolute locations of x' and x'' , not only their relative locations. This is caused by the stochasticity in the conditional expectation function.

The model above is specified up to second order, hence it can be applied in a linear, Bayesian framework. The proceeding discussion will be exact if $[Z(x); x \in A]$ and $[B_l; l = 1, \dots, L]$ are jointly Gaussian.

Recall that the two first moments of $[B_l; l = 1, \dots, L]$ are assumed known *a priori* and that it can be interpreted as a qualified guess for the coefficients in a universal kriging model for $[Z(x); x \in A]$. Qualified guess means the best guess with associated uncertainties before the observations are available. The random function of interest, $[Z(x); x \in A]$, is unknown. Assume, however, that the latter can be observed in the locations, $x_i \in A; i = 1, \dots, N$.

ESTIMATION OF SPATIAL COVARIANCE

The objective is to obtain an estimator for $C_{Z|B}(\cdot)$, based on model assumptions and set of observations $[Z(x_i); i = 1, \dots, N]$, in order to be able to solve the Bayesian kriging system.

From expression (3),

$$\begin{aligned}
\text{Cov} [Z(x'), Z(x'')] &= E[Z(x')Z(x'')] - E[Z(x')] E[Z(x'')] \\
&= C_{Z|B}(x' - x'') + \sum_l \sum_k \sigma_{lk} f^l(x') f^k(x'')
\end{aligned}$$

which entails

$$\begin{aligned}
C_{Z|B}(x' - x'') &= E[Z(x') Z(x'')] \\
&\quad - \sum_l \mu_l f^l(x') \sum_l \mu_l f^l(x'') \\
&\quad - \sum_l \sum_k \sigma_{lk} f^l(x') f^k(x'')
\end{aligned} \tag{4}$$

Let h be a vector in the reference domain A and define a set of pairs of indices D_h by

$$D_h: [(i, j) | x_i - x_j = h \text{ * or } x_j - x_i = h; i, j = 1, \dots, N]$$

$$N_h = \#D_h$$

An estimator for $C_{Z|B}(h)$ is

$$\begin{aligned}\hat{C}_{Z|B}(h) = & \frac{1}{N_h} \sum_{(i,j) \in D_h} Z(x_i)Z(x_j) - \sum_l \mu_l f^k(x_i) \sum_l \mu_l f^k(x_j) \\ & - \sum_l \sum_k \sigma_{lk} f^l(x_i) f^k(x_j)\end{aligned}$$

To show that $\hat{C}_{Z|B}(h)$ is centered around $C_{Z|B}(h)$ for all h is simple. Note in particular that a centered estimator is possible to obtain for $C_{Z|B}(0)$. This is made possible by the assumption that a prior qualified guess on the coefficients is available. As in traditional kriging theory, the spatial covariance function will be assumed to be known in the development of the kriging relations.

BAYESIAN KRIGING

Assume that the following set of observations of the random function of interest, $[Z(x); x \in A]$, is available

$$[Z(x_i); i = 1, \dots, N]$$

Because the marginal expected function of $[Z(x); x \in A]$ is known, the centered random function will be considered

$$[Z^T(x) = Z(x) - \mu_Z(x); x \in A]$$

with the corresponding set of observations

$$[Z^T(x_i) = Z(x_i) - \mu_Z(x_i); i = 1, \dots, N]$$

Consider a linear predictor for $Z(x_0)$, with x_0 an arbitrary location within A

$$Z^*(x_0) = \sum_{i=1}^N \alpha_i Z^T(x_i) + \mu_Z(x_0)$$

with α : $[\alpha_i; i = 1, \dots, N]$ a set of constant weights to be determined.

To show that the predictor is centered around $\mu_Z(x_0)$ is simple because $[Z^T(x); x \in A]$ is centered around zero.

The prediction variance is

$$\begin{aligned}\text{Var} [Z(x_0) - Z^*(x_0)] &= \text{Var} [Z(x_0)] - 2 \sum_i \alpha_i \text{Cov} [Z(x_0), Z^T(x_i)] \\ &\quad + \sum_i \sum_j \alpha_i \alpha_j \text{Cov} [Z^T(x_i), Z^T(x_j)] \\ &= C_Z(x_0, x_0) - 2 \sum_i \alpha_i C_Z(x_0, x_i) \\ &\quad + \sum_i \sum_j \alpha_i \alpha_j C_Z(x_i, x_j)\end{aligned}$$

because covariance is invariant to shifts in the variables.

From this and expression (3)

$$\begin{aligned}
 \text{Var} [Z(x_0) - Z^*(x_0)] &= C_{Z|B}(0) + \sum_l \sum_k \sigma_{lk} f^l(x_0) f^k(x_0) \\
 &\quad - 2 \sum_i \alpha_i (C_{Z|B}(x_0 - x_i) \\
 &\quad + \sum_l \sum_k \sigma_{lk} f^l(x_0) f^k(x_i)) \\
 &\quad + \sum_i \sum_j \alpha_i \alpha_j (C_{Z|B}(x_i - x_j) \\
 &\quad + \sum_l \sum_k \sigma_{lk} f^l(x_i) f^k(x_j)) \quad (5)
 \end{aligned}$$

Note that the expression cannot be written in terms of $\gamma_{Z|B}(\cdot)$ by simply substituting $C_{Z|B}(\cdot)$ by $-\gamma_{Z|B}(\cdot)$ as usually is done in universal kriging. The fact that the usual unbiasedness constraint, $\sum \alpha_i = 1$, is missing makes the substitution invalid. The same effect appears in the simple kriging expression. However, the constraint $\sum \alpha_i = 1$ may be forced on the weights and hence validate the substitution. In this case, a predictor with somewhat larger prediction variance will be obtained, but the somewhat simpler expression involving $\gamma_{Z|B}(\cdot)$ instead of $C_{Z|B}(\cdot)$ can be applied, and the expression will be valid for intrinsic random functions of order one also.

Returning to expression (5), $[f^l(x); x \in A]; l = 1, \dots, L$ are user specified and $\sigma_{lk}; l, k = 1, \dots, L$ are given *a priori*, whereas $C_{Z|B}(\cdot)$ has to be estimated from observations. A procedure for estimating this covariance function was discussed in the previous section. Proceeding as if $C_{Z|B}(\cdot)$ is known, the prediction variance is ensured to be nonnegative if $C_{Z|B}(\cdot)$ and $[\sigma_{lk}]$ are constrained to be a positive definite function and a positive definite matrix, respectively. The set of weights, α : $[\alpha_i; i = 1, \dots, N]$ will be determined by minimizing the prediction variance

$$\min_{\alpha} \text{Var} [Z(x_0) - Z^*(x_0)]$$

This corresponds to using the following Bayesian kriging system

$$\begin{aligned}
 &\sum_i \alpha_i \left(C_{Z|B}(x_i - x_j) + \sum_l \sum_k \sigma_{lk} f^l(x_i) f^k(x_j) \right) \\
 &= C_{Z|B}(x_0 - x_j) + \sum_l \sum_k \alpha_{lk} f^l(x_0) f^k(x_j); j = 1, \dots, N \quad (6)
 \end{aligned}$$

Written in matrix notation

$$(C + F \sum F^T) \cdot \alpha = c_0 + F \sum f_0$$

and the predictor is

$$Z^*(x_0) = \mu^T f_0 + (c_0 + F \sum f_0)^T (C + F \sum F^T)^{-1} (Z - F\mu) \quad (7)$$

and the prediction variance is

$$\begin{aligned} \text{Var} [Z(x_0) - Z^*(x_0)] &= (c_{00} + f_0^T \sum f_0) \\ &\quad - (c_0 + F \sum f_0)^T (C + F \sum F^T)^{-1} \\ &\quad \cdot (c_0 + F \sum f_0) \end{aligned}$$

with

$$\alpha = (\alpha_i) - N \times 1 \text{ vector}$$

$$Z = (Z_i) - N \times 1 \text{ vector with } Z_i = Z(x_i)$$

$$C = (c_{ij}) - N \times N \text{ matrix with } c_{ij} = C_{Z|B}(x_i - x_j)$$

$$\mu = (\mu_l) - L \times 1 \text{ vector}$$

$$F = (f_{il}) - N \times L \text{ matrix with } f_{il} = f^l(x_i)$$

$$\sum = (\sigma_{lk}) - L \times L \text{ matrix}$$

$$c_0 = (c_{i0}) - N \times 1 \text{ vector with } c_{i0} = C_{Z|B}(x_i - x_0)$$

$$f_0 = (f_{l0}) - L \times 1 \text{ vector with } f_{l0} = f^l(x_0)$$

$$c_{00} \quad - \text{ scalar equal } C_{Z|B}(0)$$

Note that the precision in the qualified guess of the coefficients will have influence on the weights and hence on actual predictions and prediction variances. Observe also that the kriging predictor is a linear combination of the prior qualified guess and deviations between observations and their respective prior guesses.

POSTERIOR DISTRIBUTION OF MODEL PARAMETERS

Define the linear estimators for the coefficients B_l ; $l = 1, \dots, L$, based on the observations by

$$\begin{aligned} E [B_l | Z(x_i); i = 1, \dots, N] \\ = \sum_i \beta_i^l Z^T(x_i) + \mu_l; l = 1, \dots, L \end{aligned}$$

with

β^l : $[\beta_i^l; i = 1, \dots, N]$ a set of constant weights to be determined.

The weights, β^l ; $l = 1, \dots, L$, may be determined in a similar fashion as for kriging, i.e., by solving

$$\text{Min}_{\beta^l} \text{Var} [B_l - E[B_l|Z(x_i); i = 1, \dots, N]]; l = 1, \dots, L$$

From this the estimates and corresponding estimation variances and covariances can be determined. Hence, for Gaussian random fields, the posterior distribution of the parameters are specified fully.

Written in matrix notation, the solution fulfills the following equations

$$(C + F \Sigma F^T) \beta = F \cdot \Sigma$$

and the posterior expectation of the parameters is

$$E[B|Z] = \mu + (F \Sigma)^T (C + F \Sigma F^T)^{-1} (Z - F\mu)$$

and the corresponding posterior covariance matrix is

$$\text{Cov} [B|Z] = \Sigma - (F \Sigma)^T (C + F \Sigma F^T)^{-1} F \Sigma$$

with

$$\beta = (\beta_{il}) - N \times L \text{ matrix with } \beta_{il} = \beta_i^l$$

$$B = (B_l) - L \times 1 \text{ vector}$$

and the remaining defined as above.

Observe that the posterior expectation is a linear combination of the prior qualified guess and the deviation between observations and their respective prior guesses.

SPECIAL CASES

The following special cases are considered.

Case A

Consider the case where location x_0 coincides with the location of one observation, let's say x_s with $1 \leq s \leq N$.

From expression (3), this entails

$$C_{Z|B}(x_0 - x_s) = C_{Z|B}(0)$$

$$C_Z(x_0, x_s) = C_{Z|B}(0) + \sum_l \sum_k \sigma_{lk} f^l(x_0) f^k(x_0).$$

By choosing $\alpha_s = 1$ and $\alpha_i = 0$; $i = 1, \dots, N$; $i \neq s$, from expression (5),

$$\text{Var} [Z(x_0) - Z^*(x_0)] = 0$$

which has to be a minimum because the prediction variance is constrained to be nonnegative. In addition, the solution to the system of equations is known to be unique, hence $\alpha_s = 1$ and $\alpha_i = 0$; $i = 1, \dots, N$; $i \neq s$ has to be the solution to the minimization.

This implies

$$Z^*(x_s) = Z^T(x_s) + \mu_Z(x_s) = Z(x_s)$$

Consequently, the predictor is an exact interpolator. The predicted surface runs through the observations regardless of the actual qualified guess on the coefficients.

Case B

Consider the case where $N < L$, which means that the number of observations is less than the number of parameters in the model.

In traditional universal kriging, no solution exists for this case. In Bayesian kriging, however, the qualified guess on the coefficients makes a solution possible even in this case. This can be seen easily from expression (6), because it includes no Lagrange parameters.

Experiences with universal kriging indicate that if N is slightly larger than L , large changes in the solution may occur when an additional observation is added. In Bayesian kriging, the qualified guess will tend to make the solutions more stable.

Case C

Consider the case where the user has no uncertainty about the value of the coefficients. This implies

$$\text{Cov} [B_l, B_k] = \sigma_{lk} = 0; l, k = 1, \dots, L$$

and

$$C_Z(x', x'') = C_Z(x' - x'') = C_{Z|B}(x' - x'').$$

Consequently, the Bayesian kriging system (6) appears as

$$C\alpha^T = c_0^T$$

This is identical to the simple kriging system.

Hence exact prior knowledge about coefficients in the model entails that the Bayesian kriging technique coincides with the simple kriging technique. This should not surprise geostatisticians, because the mean of the process is known and does not need to be estimated.

Case D

Consider the case where the user is not able to come up with any qualified guess on the coefficients. In the Bayesian framework, this corresponds to specifying a flat or uniform prior, which occurs when the prior covariance matrix tends to infinity and entails that all elements of Σ^{-1} tend toward zero.

Define the universal kriging predictor for a random function $[Z(x); x \in A]$ with characteristics

$$E[Z(x)] = \sum_{l=1}^L a_l f^l(x)$$

$$\text{Cov}[Z(x'), Z(x'')] = C_{Z|B}(x' - x'')$$

by $Z_U(x_0)$. Then (Appendix A),

$$Z^*(x_0) \rightarrow Z_U(x_0)$$

when $\Sigma \rightarrow \infty$ in a sense precisely defined (Appendix A).

Hence no prior knowledge about the coefficients in the model entails that the Bayesian kriging technique coincides with the familiar universal kriging technique. This should not surprise geostatisticians.

AN EXAMPLE

The example concerns depth conversion of seismic reflection times. The actual data comes from the geologic horizon outlining a North Sea petroleum reservoir. The prediction is based on the seismic reflection times as a surface and the available depth observations in wells (Fig. 1). Geophysicists have contributed with a model for seismic velocity and qualified guesses on its parameters.

A plot of depths vs. reflection times in the wells (Fig. 2) indicates the sequence in which the wells are drilled. This figure reveals the fact that preferential drilling frequently takes place in petroleum exploration. The first four wells are drilled at shallow depths which can be identified from the seismic data. Thus the first four observations are located inconveniently in the design space for estimation of the velocity function. A Bayesian procedure which allows for prior guesses on the velocity function is expected to improve the estimation by partly correcting for the effect of preferential sampling.

Let the depth to the seismic horizon be represented by the random function $[Z(x); x \in A]$, with the following properties:

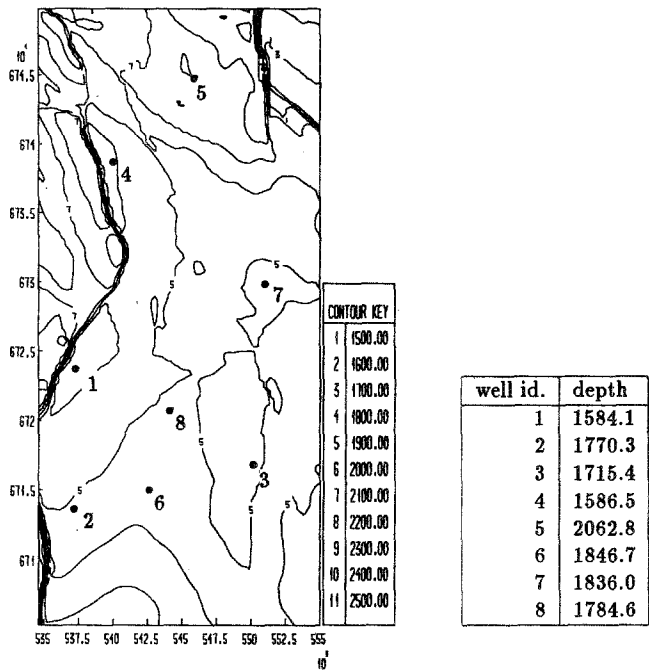


Fig. 1. Map of seismic reflection times and list of depths observed in the wells.

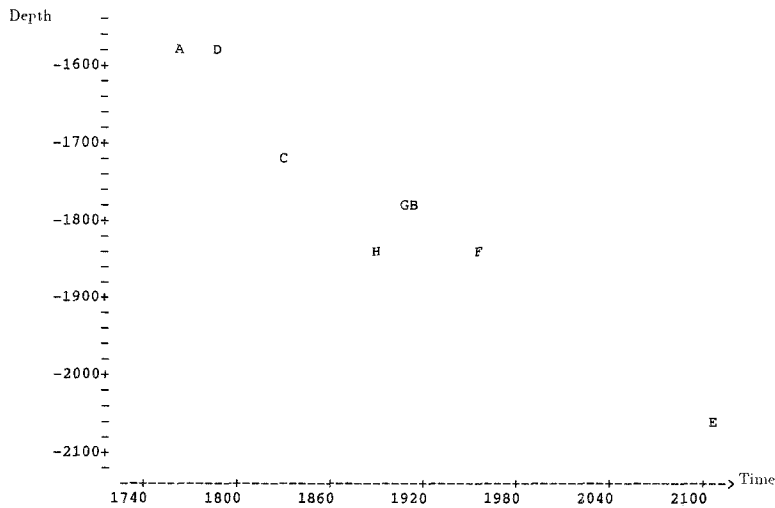


Fig. 2. Plot of depth vs. reflection time in wells, indicating the sequence of drilling.

- Expectation:

$$\begin{aligned}
 E[Z(x) | B_1, B_2, B_3] &= v(t(x), x, B_1, B_2, B_3) \cdot t(x) \\
 &= \left(B_1 + B_2 t(x) + B_3 \frac{x^1 - x_{\min}^1}{x_{\max}^1 - x_{\min}^1} \right) t(x) \\
 &= B_1 t(x) + B_2 t^2(x) + B_3 \frac{x^1 - x_{\min}^1}{x_{\max}^1 - x_{\min}^1} t(x)
 \end{aligned}$$

with $v(\cdot)$ the expected average velocity to the reflector, $[t(x); x \in A]$ the seismic reflection time surface considered as known, $x = [x^1, x^2]$ the reference location, $x^1, x_{\min}^1, x_{\max}^1$ the east-west coordinate of the location and the corresponding minimum and maximum values of area A, and B the vector of random variables constituting the coefficients in the velocity relation.

Note that the conditional expectation appears as a linear combination of known functions with random coefficients. The first term represents the traditional velocity $\times$ time relation, the second represents increasing velocity with time, and the third is due to varying velocity in the east-west direction of the reservoir. The latter is caused by an east-west thinning geologic layer with extreme velocity properties in the overburden.

- Spatial covariance:

$$\text{Cov}[Z(x'), Z(x'') | B_1, B_2, B_3] = 200 \times \text{Sph}(x' - x'' | 2800)$$

with

$$\text{Sph}(x | x_R) = \frac{3}{2} \cdot \frac{x}{x_R} - \frac{1}{2} \left(\frac{x}{x_R} \right)^3$$

Parameter values in the spatial covariance expression are specified based on previous experience.

The user has specified the following prior distribution on the parameters B

$$\begin{aligned}
 E[B] &= \begin{bmatrix} 6.0 \times 10^{-1} \\ 2.0 \times 10^{-4} \\ 7.0 \times 10^{-2} \end{bmatrix} \\
 \text{Cov}[B] &= \theta^2 \times \begin{bmatrix} 3.0 \times 10^{-3} & 0 & 0 \\ 0 & 1.0 \times 10^{-8} & 0 \\ 0 & 0 & 5.0 \times 10^{-4} \end{bmatrix}
 \end{aligned}$$

with θ as a scaling factor to be varied in the example. These distributions are obtained from experience from other field studies. This model specification corresponds to the Bayesian approach to kriging and the consequences will be studied in the following two cases.

Case A: Varying Precision in Prior Specification of B

The number of wells are kept constant to three, the first three listed (Fig. 1). The precision in the qualified guess is varied through θ . The consequences on the posterior parameter distribution and the resulting map are presented (Table 1 and Fig. 3).

The posterior distribution of the parameters as a function of θ is specified (Table 1). Recall that $\theta = 0.0$ corresponds to simple kriging. Hence, no uncertainty exists in the parameter specification, and the parameters remain unchanged. At the other extreme, $\theta = \infty$ corresponds to universal kriging. Hence, no prior knowledge about the parameters is available, and they are determined

Table 1. Posterior Distribution of Parameters and Velocity at a Location on the Center of the Map [with $t(x) = 1900.0$] When Varying the Precision in the Prior A^a

θ	B_1	B_2	B_3	$V(x_0)$
0.0	0.6000 0 0 0	0.0002 0 0 0	0.0700 0 0 0	1.015
0.01	0.5991 $0.296 \cdot 10^{-6}$ $-0.236 \cdot 10^{-10}$ $-0.217 \cdot 10^{-9}$	0.00019 $-0.236 \cdot 10^{-10}$ $0.855 \cdot 10^{-12}$ $-0.133 \cdot 10^{-11}$	0.0699 $-0.217 \cdot 10^{-9}$ $-0.133 \cdot 10^{-11}$ $0.500 \cdot 10^{-7}$	1.003
0.1	0.5943 $0.277 \cdot 10^{-4}$ $-0.141 \cdot 10^{-7}$ $-0.137 \cdot 10^{-6}$	0.00017 $-0.141 \cdot 10^{-7}$ $0.129 \cdot 10^{-10}$ $-0.773 \cdot 10^{-9}$	0.0691 $-0.137 \cdot 10^{-6}$ $-0.773 \cdot 10^{-9}$ $0.488 \cdot 10^{-5}$	0.944
1.0	0.5865 $0.241 \cdot 10^{-2}$ $-0.130 \cdot 10^{-5}$ $-0.309 \cdot 10^{-4}$	0.00017 $-0.130 \cdot 10^{-5}$ $0.711 \cdot 10^{-9}$ $-0.111 \cdot 10^{-7}$	0.0501 $-0.309 \cdot 10^{-4}$ $-0.111 \cdot 10^{-7}$ $0.152 \cdot 10^{-3}$	0.938
10.0	0.5460 $0.178 \cdot 10^{-1}$ $-0.958 \cdot 10^{-5}$ $-0.285 \cdot 10^{-3}$	0.0002 $-0.958 \cdot 10^{-5}$ $0.518 \cdot 10^{-8}$ $0.113 \cdot 10^{-6}$	0.0423 $-0.285 \cdot 10^{-3}$ $0.113 \cdot 10^{-6}$ $0.223 \cdot 10^{-3}$	0.938
∞	0.5428 $0.190 \cdot 10^{-1}$ $-0.102 \cdot 10^{-4}$ $-0.309 \cdot 10^{-3}$	0.0002 $-0.102 \cdot 10^{-4}$ $0.553 \cdot 10^{-8}$ $0.127 \cdot 10^{-6}$	0.0422 $-0.309 \cdot 10^{-3}$ $0.127 \cdot 10^{-6}$ $0.223 \cdot 10^{-3}$	0.938

^aBelow the posterior expectation of the parameters is written the posterior covariance matrix.

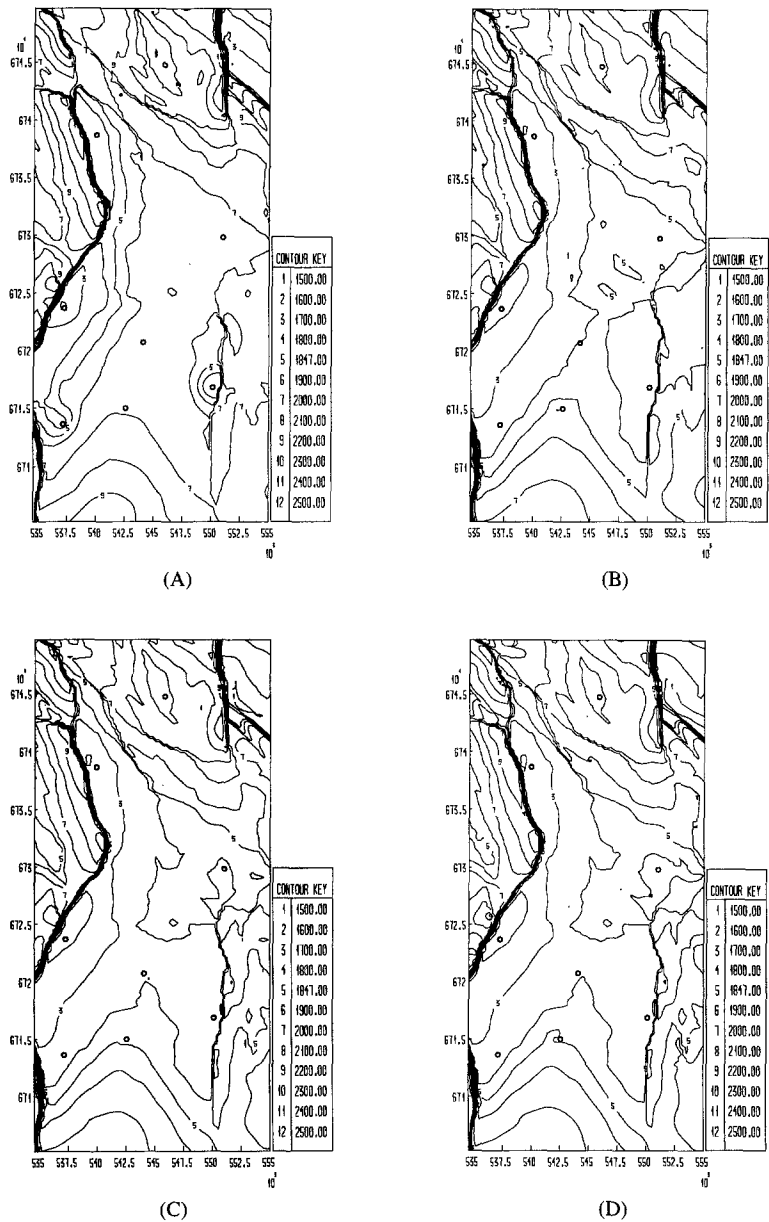


Fig. 3. Kriging maps in Case A. (A) $\theta = 0.0$; (B) $\theta = 0.1$; (C) $\theta = 1.0$; (D) $\theta = \infty$.

solely from available observations. Parameter values converge toward the universal kriging solution with increasing θ (Table 1). Note that the convergence is not monotone in the parameters.

For $\theta = 0$, the simple kriging solution, the contoured surface around observations (Fig. 3) appears like cones due to conditioning. This is caused by no uncertainty in the parameters, hence no flexibility in the model. With increasing θ and large uncertainty in the model specification, the model adapts better to observations. In the extreme case $\theta = \infty$, corresponding to universal kriging, no prior knowledge about the parameters are specified, hence the posterior model is defined by observations only.

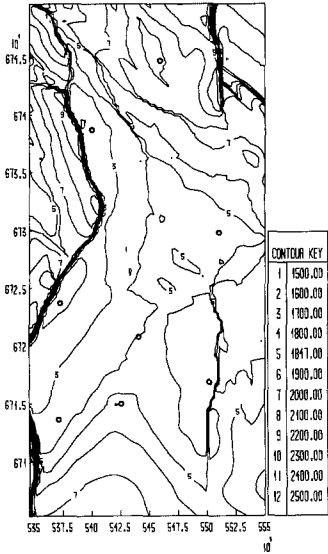
Case B: Varying Number of Wells

Here, precision in the qualified guess of the parameters is kept constant, $\theta = 1.0$, but the number of wells is varied. Numbers used are one, three, five, and eight in the sequence listed (Fig. 1). The consequences on the posterior distribution of parameters and the resulting maps are presented (Table 2 and Fig. 4).

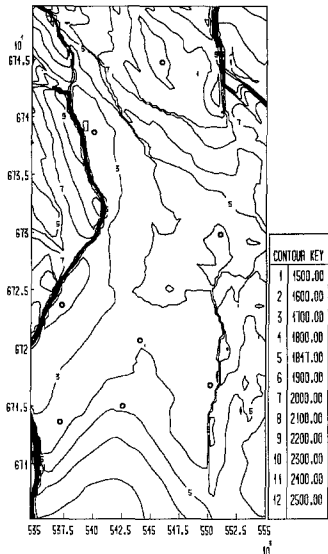
Table 2. Posterior Distribution of Parameters and Velocity at a Location on the Center of the Map [with $t(x) = 1900.0$], When Varying the Number of Wells Used in the Conditioning^a

No. of wells	B_1	B_2	B_3	$V(x_0)$
1	0.5942	0.00017	0.0699	0.944
	$0.274 \cdot 10^{-2}$	$-0.155 \cdot 10^{-5}$	$-0.572 \cdot 10^{-5}$	
	$-0.155 \cdot 10^{-5}$	$0.895 \cdot 10^{-9}$	$-0.337 \cdot 10^{-7}$	
	$-0.572 \cdot 10^{-5}$	$-0.337 \cdot 10^{-7}$	$0.500 \cdot 10^{-3}$	
3	0.5865	0.00017	0.0501	0.938
	$0.241 \cdot 10^{-2}$	$-0.130 \cdot 10^{-5}$	$-0.309 \cdot 10^{-4}$	
	$-0.130 \cdot 10^{-5}$	$0.711 \cdot 10^{-9}$	$-0.111 \cdot 10^{-7}$	
	$-0.309 \cdot 10^{-4}$	$-0.111 \cdot 10^{-7}$	$0.152 \cdot 10^{-3}$	
5	0.5581	0.00018	0.0532	0.936
	$0.131 \cdot 10^{-2}$	$-0.701 \cdot 10^{-6}$	$0.679 \cdot 10^{-4}$	
	$-0.701 \cdot 10^{-6}$	$0.385 \cdot 10^{-9}$	$0.679 \cdot 10^{-7}$	
	$0.679 \cdot 10^{-4}$	$0.645 \cdot 10^{-7}$	$0.143 \cdot 10^{-3}$	
8	0.5654	0.00018	0.0663	0.940
	$0.127 \cdot 10^{-2}$	$-0.674 \cdot 10^{-6}$	$0.362 \cdot 10^{-4}$	
	$-0.674 \cdot 10^{-6}$	$0.364 \cdot 10^{-9}$	$-0.427 \cdot 10^{-7}$	
	$0.362 \cdot 10^{-4}$	$-0.427 \cdot 10^{-7}$	$0.102 \cdot 10^{-3}$	

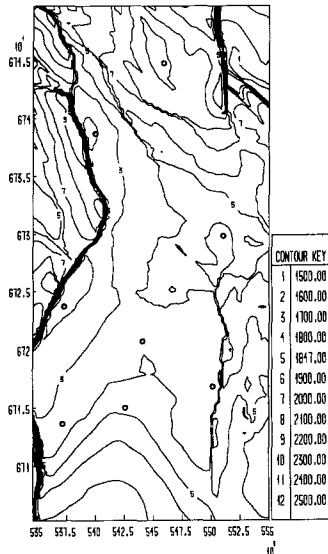
^aBelow the posterior expectation of the parameters is written the posterior covariance matrix.



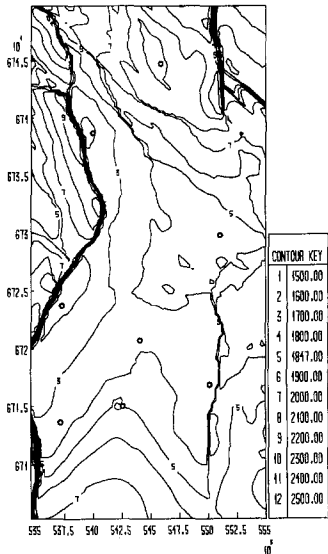
(A)



(B)



(C)



(D)

Fig. 4. Kriging maps, posterior velocity map, and prediction variance map in Case B. (A) Kriging map/one well; (B) Kriging map/three wells; (C) Kriging map/five wells; (D) Kriging map/eight wells; (E) posterior velocity map/three wells; (F) prediction variance map/three wells.

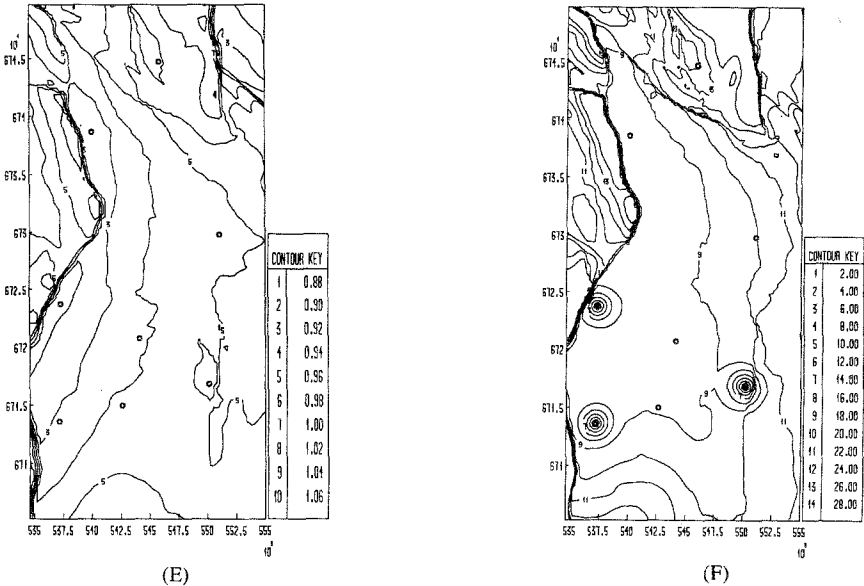


Fig. 4. Continued.

A kriging solution exists with only one well present (Table 2). Recall that three parameters are defined in the model. The prior distribution on the parameters makes it possible to obtain a solution even when the number of wells is less than three. The posterior parameter values will change, of course, with both number and values of wells. The posterior variances of the parameters will decline monotonically with the number of wells as expected. Note, however, although no correlation between parameters was specified *a priori*, the posterior parameters appear correlated. The posterior estimators all draw on the information of the same observations, hence it makes sense intuitively that they are correlated.

Well observations have influence on the results in two ways (Fig. 4)—first, globally through the posterior update of parameters; secondly, locally through the conditioning on the well observations. The result constitutes a trade-off between the prior guess and the observations weighted according to respective variance values. This can be observed (Fig. 4A–D) by recognizing changes both far away from and close to observations that are introduced.

The predicted average velocity, $v(x)$ (Fig. 4E), based on the kriging map for three wells and $\theta = 1.0$ (Fig. 4B), is defined by dividing the kriging map by the seismic reflection time map. Recall that the velocity was assumed dependent on both reflection time and east–west location. The former dependence can be observed from changing velocity over the discontinuities in the reflection time map, the latter from the overall east–west trend.

The prediction variance of the kriging map (Fig. 4F) for three wells and $\theta = 1.0$ (Fig. 4B), indicates that the range of spatial covariance is such that none of the observations are correlated in this example. Note also that the precision varies over the area due to the dependence on the reflection times and the east–west location. The precision will be influenced by both the prior specified precision in the model parameters, the spatial covariance function, and the available observations.

CLOSING REMARKS

A methodological base for using prior guesses in evaluation of spatial phenomena is developed. The Bayesian approach to kriging is bridging the gap between simple and universal kriging in geostatistical terminology. The method is expected to have a great potential for use in applications with the following characteristics:

- users have considerable experience and knowledge about the phenomenon to be evaluated;
- only a small number of observations is available; and
- variables related to the phenomenon are known exhaustively, which allows for a regression formulation, but which also encourages preferential sampling.

Applications such as the evaluation of petroleum reservoirs, geotechnical analysis, and hydrological mapping, often exhibit these characteristics.

The kriging technique is simple to implement on a computer as an extension of an existing kriging program. The fact that a unified expression for simple and universal kriging is defined makes the program maintenance even more convenient. A methodological extension to include conditional simulation is expected to be simple. However, the final test for usefulness of the method will be through future application to real problems.

APPENDIX A: CONSEQUENCES OF FLAT PRIOR DISTRIBUTIONS

Recall the expression for the Bayesian kriging predictor,

$$Z^*(x_0) = \mu^T f_0 + \left(c_0 + F \sum f_0 \right)^T \left(C + F \sum F^T \right)^{-1} (Z - F\mu)$$

and for the traditional universal kriging predictor

$$Z_U(x_0) = B_U^T f_0 + c_0^T C^{-1} (Z - FB_U)$$

where B_U is the generalized least-squares estimator of the parameters:

$$\begin{aligned} B_U &= (F^T C^{-1} F)^{-1} F^T C^{-1} Z \\ &= H F^T C^{-1} Z \end{aligned}$$

with

$$H = (F^T C^{-1} F)^{-1}$$

In the demonstration, the following matrix identity, which is shown in Mardia *et al.* (1979), is needed:

$$(C + F \Sigma F^T)^{-1} = C^{-1} - C^{-1} F G_{\Sigma} F^T C^{-1}$$

with

$$G_{\Sigma} = (H^{-1} + \Sigma^{-1})^{-1}$$

Note that $G_{\Sigma} \rightarrow H$ when $\Sigma \rightarrow \infty$, in the sense that the smallest eigenvalue of Σ converges to infinity.

By applying this identity one may show:

$$\begin{aligned} Z^*(x_0) &= \mu^T f_0 + f_0^T \Sigma (I - H^{-1} G_{\Sigma}) F^T C^{-1} (Z - F \mu) \\ &\quad + c_0^T C^{-1} Z + c_0^T C^{-1} F (G_{\Sigma} H^{-1} - I) \mu \\ &\quad - c_0^T C^{-1} F G_{\Sigma} F^T C^{-1} Z \\ &= \mu^T f_0 + f_0^T \Sigma (I - H^{-1} G_{\Sigma}) H^{-1} B_U \\ &\quad - f_0^T \Sigma (I - H^{-1} G_{\Sigma}) F^T C^{-1} F \mu \\ &\quad + c_0^T C^{-1} (Z - F G_{\Sigma} H^{-1} B_U) \\ &\quad + c_0^T C^{-1} F (G_{\Sigma} H^{-1} - I) \mu \end{aligned}$$

By using the identity above one may show,

$$(I - H^{-1} G_{\Sigma}) = \Sigma^{-1} G_{\Sigma}$$

and hence:

$$\begin{aligned} Z^*(x_0) &= \mu^T f_0 + f_0^T G_{\Sigma} H^{-1} B_U \\ &\quad - f_0^T G_{\Sigma} H^{-1} \mu \\ &\quad + c_0^T C^{-1} F (G_{\Sigma} H^{-1} - I) \mu \\ &\quad + c_0^T C^{-1} (Z - F G_{\Sigma} H^{-1} B_U) \end{aligned}$$

Let $\Sigma \rightarrow \infty$, which implies $G_\Sigma \rightarrow H$, then

$$Z^*(x_0) \rightarrow f_0^T B_U + c_0^T C^{-1}(Z - F B_U) = Z_U(x_0)$$

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